Week 4 - Tagging strategy, Tag policies, Resource Groups

DOP-C02 Domains 2 and 6. Typical questions:

1. **Every resource** must have `CostCenter` / `Owner` / `Environment`, and **creation without them must be blocked** = **SCP / IAM** `aws:RequestTag`.
2. Tag **keys and values consistent** (case, allowed values) **across the organization** = **tag policies**.
3. Finance needs **costs per team / project** = **cost allocation tags** (activated in the **management account**) + Cost Categories.
4. Find / fix **untagged or wrongly tagged** resources = **Config** `required-tags`, **Tag Editor**, tag policy **compliance report**, EventBridge + Lambda auto-tagging.
5. Run commands / patch / monitor **resources defined by tags** = **Resource Groups** / tag-based targets.
6. Permissions that **scale with tags** = **ABAC** (`aws:PrincipalTag` = `aws:ResourceTag`).

The big idea: **tags are metadata (key = value) that drive cost, automation, security and operations**. Define a **tag dictionary**, then enforce in layers: **set at creation (IaC) -> prevent bad creation (SCP / IAM) -> standardise (tag policy) -> detect (Config) -> fix (remediation / Tag Editor)**.

Part 1 - Tag basics

Fact	Detail
Format	**Key = Value**; both **case-sensitive** (`CostCenter` is not `costcenter`)
Limits	Up to **50 user-defined tags** per resource; key up to **128** chars; value up to **256** chars
Reserved prefix	`aws:` = AWS-generated (`aws:cloudformation:stack-name`, `aws:createdBy`, `aws:servicecatalog:*`); can't edit / delete; don't count toward 50
Scope	Per resource, per Region; **S3 bucket tags are not S3 object tags**
Not secrets	**Never put sensitive data in tags** (billing, console, APIs)
Cost	Tagging is free

Part 2 - Designing a tagging strategy

Category	Example keys	Used for
Technical	`Name`, `Environment` (dev / test / prod), `Application`, `Component`, `Version`	Operations, Resource Groups, dashboards
Business / cost	`CostCenter`, `Owner`, `Project`, `BusinessUnit`, `Customer`	**Cost allocation**, chargeback / showback
Security / compliance	`DataClassification`, `Compliance` (PCI / HIPAA)	**ABAC**, SCP / IAM conditions, Macie scoping, audits
Automation	`Schedule`, `Backup`, `Patch Group`, `AutoShutdown`	**Instance Scheduler**, **AWS Backup** tag selection, **Patch Manager**, Lambda jobs

Best practices:

1. **Small mandatory set** (`CostCenter`, `Owner`, `Environment`, `Application`) + optional tags, documented in a **tagging dictionary** (key, allowed values, owner, purpose).
2. **One naming convention**: PascalCase (`CostCenter`) or lowercase-hyphen (`cost-center`); optional **company prefix** (`acme:cost-center`).
3. **Allowed value lists** (`prod`, `staging`, `dev` - not `Prod` / `production` / `PRD`).
4. **Automate tagging at creation** - don't rely on people.
5. **Enforce in layers** and **measure compliance**.
6. **Accounts first, tags second**: separate accounts give hard boundaries and easy cost split; tags add detail inside accounts.
7. **Cost allocation tags apply from activation onward** (limited backfill) - tag early.

Part 3 - Applying tags automatically

Method	Detail
CloudFormation stack tags	Stack tags **propagate to supported resources** (+ automatic `aws:cloudformation:*` tags); `--tags` on create / deploy; StackSets support them
CDK	`Tags.of(app).add('CostCenter', '100')` tags every taggable construct (an Aspect)
SAM	`Tags` on functions / `Globals`, `sam deploy --tags`
Terraform	Provider `default_tags`
Service Catalog	**TagOptions** + automatic `aws:servicecatalog:*` tags
EC2 launch templates	`TagSpecifications` for instance, volume, ENI, spot request
Auto Scaling	ASG tags with `PropagateAtLaunch: true`
ECS	`propagateTags` (`TASK_DEFINITION` / `SERVICE`) + `enableECSManagedTags`
Snapshots / AMIs / Backup	Copy-tags options (`CopyTagsToSnapshot`, Backup copies tags to recovery points)
Auto-tag the creator	**EventBridge on CloudTrail** `Create*` **/** `RunInstances` **-> Lambda adds** `CreatedBy`; AWS-generated `aws:createdBy` cost allocation tag

Part 4 - Preventing untagged or invalid creation (SCP / IAM)

Condition key	Meaning	Typical use
`aws:RequestTag/<key>`	Tag **in the create / tag request**	**Require** a tag, **restrict values**
`aws:TagKeys`	**List of tag keys** in the request	**Allow only approved keys** (`ForAllValues:StringEquals`), **block removal** of keys
`aws:ResourceTag/<key>`	Tag **already on the resource**	**ABAC**, protect tagged resources
`aws:PrincipalTag/<key>`	Tag **on the calling principal / session**	**ABAC**

Require CostCenter when creating instances and volumes

```
{ "Sid": "RequireCostCenter",
  "Effect": "Deny",
  "Action": ["ec2:RunInstances", "ec2:CreateVolume"],
  "Resource": ["arn:aws:ec2:*:*:instance/*", "arn:aws:ec2:*:*:volume/*"],
  "Condition": { "Null": { "aws:RequestTag/CostCenter": "true" } } }
```

Restrict allowed values

```
{ "Effect": "Deny", "Action": "ec2:RunInstances",
  "Resource": "arn:aws:ec2:*:*:instance/*",
  "Condition": {
    "StringNotEquals": { "aws:RequestTag/Environment": ["dev", "test", "prod"] } } }
```

Prevent removing or changing mandatory tags

```
{ "Effect": "Deny",
  "Action": ["ec2:DeleteTags", "ec2:CreateTags"],
  "Resource": "*",
  "Condition": {
    "ForAnyValue:StringEquals": { "aws:TagKeys": ["CostCenter", "Owner"] },
    "ArnNotLike": { "aws:PrincipalARN": "arn:aws:iam::*:role/TagAdminRole" } } }
```

- **Tag-on-create** must be supported by the service / API (EC2 `RunInstances` with `TagSpecifications`); **not every action supports** `aws:RequestTag`.
- **Tag policies don't make tags mandatory - SCP / IAM with** `aws:RequestTag` **does.**

Part 5 - Tag policies (AWS Organizations)

Standardise **key capitalisation**, **allowed values**, and optional **enforcement** for resource types.

```
{ "tags": {
    "costcenter": {
      "tag_key":      { "@@assign": "CostCenter" },
      "tag_value":    { "@@assign": ["100", "200", "300*"] },
      "enforced_for": { "@@assign": ["ec2:instance", "ec2:volume", "s3:bucket"] } },
    "environment": {
      "tag_key":   { "@@assign": "Environment" },
      "tag_value": { "@@assign": ["dev", "test", "prod"] } } } }
```

Feature	Detail
Enable	Organizations (**all features**), enable **TAG_POLICY** on the root, attach to root / OU / account
`tag_key`	**Exact capitalisation**; `costcenter` on a resource = **non-compliant**
`tag_value`	**Allowed values** (trailing `*` wildcard)
`enforced_for`	**Non-compliant tagging operations blocked** for listed types (not all types supported)
Not done	**Doesn't require the tag**; **untagged resource = compliant**; **doesn't fix existing** tags
Inheritance	Root -> OU -> account merge into an **effective tag policy**; `@@assign` (overwrite), `@@append` (add values), `@@remove` (remove); `@@operators_allowed_for_child_policies` limits children
Compliance	Effective policy + non-compliant resources per account (Tag Editor, `GetComplianceSummary`); **org-wide report** from the **management account** (`StartReportCreation`) to **S3 in us-east-1**

Need	Tool
Correct **case and allowed values** org-wide (block bad values on enforced types)	**Tag policy**
Tag must be present at creation	**SCP / IAM** `aws:RequestTag` **(Null)**
Report missing tags	**Config** `required-tags`
Fix missing tags	Config remediation (custom SSM) / Lambda / Tag Editor

Part 6 - Detecting and fixing tag problems

Tool	Use
Config `required-tags`	Up to **6 keys**, optional values, scoped by type; org-wide via **organization conformance pack**, view in **aggregator**
Tag Editor	**Search across Regions** by type and tag (incl. **not tagged**); **bulk add / edit / remove**; CSV export; tag policy compliance
Resource Groups Tagging API	`GetResources`, `TagResources` / `UntagResources` (bulk, cross-service), `GetTagKeys`, `GetTagValues`, `GetComplianceSummary`, `StartReportCreation`
Cost Explorer / CUR	Group cost by tag; **No tag value** = untagged spend
Remediation	Config remediation with **custom SSM Automation**; **EventBridge + Lambda** auto-tagging; Tag Editor bulk fixes
Notifications	EventBridge `Tag Change on Resource` events or Config compliance changes

Part 7 - Cost allocation tags

Fact	Detail
Activation	**User-defined tags activated in the management (payer) account** (Billing -> Cost allocation tags)
Timing	**Up to 24 hours**; applies **from activation onward** (limited backfill)
AWS-generated	`aws:createdBy` - also activated in Billing
Where	**Cost Explorer**, **Budgets** (filter by tag), **CUR / Data Exports**
Cost Categories	Rules mapping **accounts / tags / services** to business categories - chargeback when tagging is incomplete
Untaggable costs	Some data transfer, support: use **account structure + Cost Categories**

"Tags exist but Cost Explorer can't group by them" = **activate cost allocation tags in the management account** (wait up to 24 hours).

Part 8 - ABAC with tags

```
{ "Effect": "Allow",
  "Action": ["ec2:StartInstances", "ec2:StopInstances"],
  "Resource": "arn:aws:ec2:*:*:instance/*",
  "Condition": {
    "StringEquals": { "aws:ResourceTag/Team": "${aws:PrincipalTag/Team}" } } }
```

- Principal tagged `Team=payments` (IAM role tag or **session tag** from Identity Center / SAML) manages only resources tagged `Team=payments`.
- **Scales**: new teams / resources need **no policy changes**, only tags (RBAC needs policies per role).
- Required: **require the tag on create** (`aws:RequestTag/Team` = `${aws:PrincipalTag/Team}`) and **deny changing the Team tag** (`aws:TagKeys`).
- Identity Center: **attributes for access control** -> session tags.
- Not every action supports resource tags in authorization.

Part 9 - Resource Groups

Named **collection of resources in one Region** managed and automated **as a unit**.

Group type	Definition	Notes
Tag-based	**Query**: resource types + **tag filters** (`Application=orders`, `Environment=prod`)	**Dynamic** - new tagged resources join automatically
CloudFormation stack-based	Supported resources in a **specific stack**	Follows the stack
Service-configured	Groups with a **service configuration** (EC2 capacity reservation groups, License Manager host resource groups)	Used by that service

Consumer	Use
Systems Manager	**Run Command, Automation, State Manager, maintenance windows, Patch Manager** target a resource group; **Application Manager**
CloudWatch	**Application Insights**, automatic dashboards per group
AWS Backup	Selects by **tags directly** (same idea)
Console	One view of an app's resources
EventBridge	**Group lifecycle events** (membership changes)

- **Groups are regional**; Tag Editor for cross-Region search.
- **AppRegistry / myApplications** (`awsApplication` tag) organise resources by application for cost and ops views.
- "Patch / run a command on all prod web servers of app X, including new ones" = **tag-based resource group (or tag targets) as SSM target**.

Part 10 - Choosing the right tool

Requirement	Best choice
Consistent tags at creation	**Stack tags / CDK** `Tags.of` **/ Terraform** `default_tags`, launch templates, ASG `PropagateAtLaunch`, ECS `propagateTags`
Block creation without tags	**SCP / IAM** `aws:RequestTag` **+** `Null`
Only approved keys	`aws:TagKeys` **with** `ForAllValues`
Prevent removing / changing key tags	**Deny** `DeleteTags` **/** `CreateTags` **for those** `aws:TagKeys`
Standardise case / values org-wide	**Tag policy** (+ `enforced_for`)
Report missing tags	**Config** `required-tags` (+ org pack + aggregator)
Find and bulk-fix across Regions	**Tag Editor** / Tagging API
Auto-tag the creator	**EventBridge -> Lambda** or `aws:createdBy`
Cost per team	**Cost allocation tags (management account)** + Cost Categories
Permissions by team	**ABAC**
Operate a set of resources	**Resource Groups**
Service Catalog launches	**TagOptions**

Part 11 - Exam traps

Trap	Truth
Tag policy makes tags mandatory	**No**; untagged = compliant; **SCP / IAM** `aws:RequestTag` **Null**
Tag policy fixes existing tags	**No**; reports, blocks bad **tag operations** on enforced types
`costcenter` vs `CostCenter`	**Case-sensitive**; `tag_key` sets the case
Tag policies on billing-only org	**All features** + TAG_POLICY enabled
Org-wide tag compliance report	**Management account**, **S3 in us-east-1**
Tags missing in Cost Explorer	**Activate cost allocation tags in management account**
Member account activates cost allocation tags	**Only the management account**
Charge back untaggable costs	**Cost Categories** + accounts
Require tag on unsupported API	Not all actions support `aws:RequestTag`; Config + remediation
Users remove CostCenter	**Deny** `DeleteTags` **/** `UntagResource` **with** `aws:TagKeys`
ABAC users re-tag to gain access	**Deny tag changes** + require on create
ASG instances miss tags	`PropagateAtLaunch` / launch template `TagSpecifications`
ECS tasks untagged	`propagateTags`
CloudFormation resources untagged	**Stack-level tags**
CDK tag everything	`Tags.of(scope).add()`
Find untagged resources in all Regions	**Tag Editor** (or Config + aggregator)
Report missing tags across accounts	**Config** `required-tags` org pack + aggregator
Auto-tag creator	**EventBridge + Lambda** / `aws:createdBy`
Patch all prod servers of app X	**Tag-based resource group** / tag targets
Resource group across Regions	**Regional**
Secrets in tags	**Never**
Edit `aws:` tags	**Can't**

Part 12 - Hands-on lab (about 60 minutes, very low cost)

Lab 1 - Tags at creation

1. Deploy a stack (SQS queue + S3 bucket) with stack tags:

```
aws cloudformation deploy --template-file app.yaml --stack-name tag-lab \
  --tags CostCenter=100 Environment=dev Application=orders
```

2. Launch template with **TagSpecifications** (instance + volume `Application=orders`) and ASG tag `Owner=team-a` with **PropagateAtLaunch**; launch one instance - instance and volume carry the tags.

Lab 2 - Require tags (IAM or SCP)

1. Attach the Part 4 policies to a test role (or as an SCP on a sandbox OU).
2. `run-instances` without tags: **denied**; with tags on instance and volume:

```
aws ec2 run-instances --image-id <ami> --instance-type t3.micro \
  --tag-specifications \
  'ResourceType=instance,Tags=[{Key=CostCenter,Value=100},{Key=Environment,Value=dev}]' \
  'ResourceType=volume,Tags=[{Key=CostCenter,Value=100},{Key=Environment,Value=dev}]'
```

3. `Environment=Prod`: **denied** (case and value). Removing `CostCenter`: **denied**.

Lab 3 - Tag policy (test org)

1. Enable **tag policies**, attach the Part 5 policy to a sandbox OU, view the **effective policy**.
2. `costcenter=100` (wrong case) = **non-compliant** in Tag Editor; `CostCenter=999` on EC2 = **blocked**; untagged instance = **allowed**.
3. Management account: **Generate compliance report** to S3 in us-east-1.

Lab 4 - Detect and fix

1. Config `required-tags` (`CostCenter`, `Owner`) for EC2 and S3: see non-compliant resources.
2. **Tag Editor**: all Regions, `Owner` = not tagged; **bulk add** `Owner=platform`; rule turns compliant.
3. Optional auto-tagging: EventBridge (`aws.ec2`, CloudTrail `RunInstances`) -> Lambda `CreateTags` with `CreatedBy = detail.userIdentity.arn`.

Lab 5 - Cost allocation tags and ABAC

1. Management account: activate `CostCenter`, `Environment`, `Application`, `aws:createdBy`; after about 24 hours group Cost Explorer by `CostCenter`.
2. ABAC: role `Dev-Payments` tagged `Team=payments` with the Part 8 policy; instance A `Team=payments`, B `Team=search`; the role can **stop A**, **not B**.

Lab 6 - Resource Groups and SSM

1. **Tag-based group** `orders-dev` (`AWS::EC2::Instance`, `AWS::S3::Bucket`, `Application=orders`, `Environment=dev`).
2. **Run Command** `AWS-RunShellScript` targeting the group; launch another tagged instance - it joins automatically.

Cleanup: delete stack, ASG, launch template, instances, policies, tag policy, Config rule, Lambda / EventBridge rule, resource group.

Part 13 - Check yourself

Questions

1. Tag limits and case rules; what's special about `aws:`?
2. Four categories for a tagging dictionary, with examples?
3. Three ways to apply tags automatically at creation?
4. Block EC2 creation without CostCenter?
5. Allow only approved keys, and prevent removing Owner?
6. What does a tag policy control, and what does `enforced_for` do?
7. Does a tag policy make a tag mandatory? Is an untagged resource compliant?
8. Tag policy inheritance and `@@assign` / `@@append` / `@@remove`?
9. Where and how to get an org-wide tag compliance report?
10. Report resources missing required tags across all accounts?
11. Find untagged resources across Regions and fix in bulk?
12. Tags exist but Cost Explorer can't group by them - why?
13. How does ABAC work, and which two extra controls are needed?
14. Types of resource groups; are they regional?
15. Patch all prod web servers of app X, including future ones?

Answers

1. **50** user tags, key **128**, value **256**, **case-sensitive**; `aws:` = AWS-generated, read-only, not counted.
2. **Technical** (Name, Environment, Application), **business** (CostCenter, Owner, Project), **security** (DataClassification, Compliance), **automation** (Schedule, Backup, Patch Group).
3. Stack tags / CDK `Tags.of` / Terraform `default_tags`; launch template TagSpecifications; ASG PropagateAtLaunch; ECS propagateTags; TagOptions.
4. **SCP / IAM Deny** RunInstances (instance / volume) with `Null aws:RequestTag/CostCenter = true`.
5. `aws:TagKeys` **+** `ForAllValues:StringEquals`; **deny DeleteTags / UntagResource** (and CreateTags) when `aws:TagKeys` contains Owner (admin exception).
6. **Key case + allowed values** org-wide; `enforced_for` blocks non-compliant tag operations for listed types.
7. **No**; **yes, compliant**; use an **SCP** `aws:RequestTag`.
8. Root -> OU -> account merge into the **effective policy**; overwrite / add / remove values; `@@operators_allowed_for_child_policies`.
9. **Management account** (`StartReportCreation`), **S3 in us-east-1**.
10. **Config** `required-tags` via **organization conformance pack** + **aggregator**.
11. **Tag Editor** (not tagged, bulk edit) or the **Tagging API**.
12. Not **activated as cost allocation tags in the management account** (up to 24h, forward).
13. Allow when `aws:ResourceTag/Team` **=** `${aws:PrincipalTag/Team}`; **require tag on create** + **deny tag changes**.
14. **Tag-based** (dynamic), **stack-based**, **service-configured**; **regional**.
15. **Tag-based resource group** (or tag targets) as the SSM target.

Quick memory summary

- **Tags** = case-sensitive key / value (50, `aws:` reserved, no secrets); small mandatory set + dictionary + naming convention + allowed values; accounts first.
- **Apply automatically**: stack tags, CDK `Tags.of`, Terraform `default_tags`, launch templates, ASG PropagateAtLaunch, ECS propagateTags, TagOptions, auto-tag creator.
- **Prevent**: `aws:RequestTag` (Null = required, values), `aws:TagKeys` (keys / no removal), `aws:ResourceTag` **/** `aws:PrincipalTag` (ABAC).
- **Tag policies**: case + values, `enforced_for`, inheritance operators, report (management -> S3 us-east-1); **don't require tags**.
- **Detect / fix**: Config `required-tags`, **Tag Editor**, Tagging API, remediation.
- **Cost**: **activate cost allocation tags in management account** + Cost Categories.
- **Resource Groups**: tag-based (dynamic) / stack-based / service-configured, **regional**; SSM targets, Application Insights.